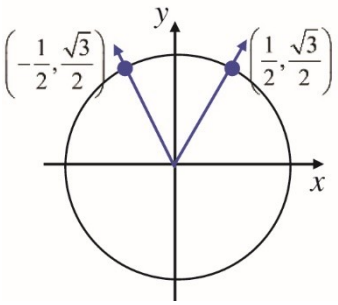
Solve Trigonometric Equations in Linear Form

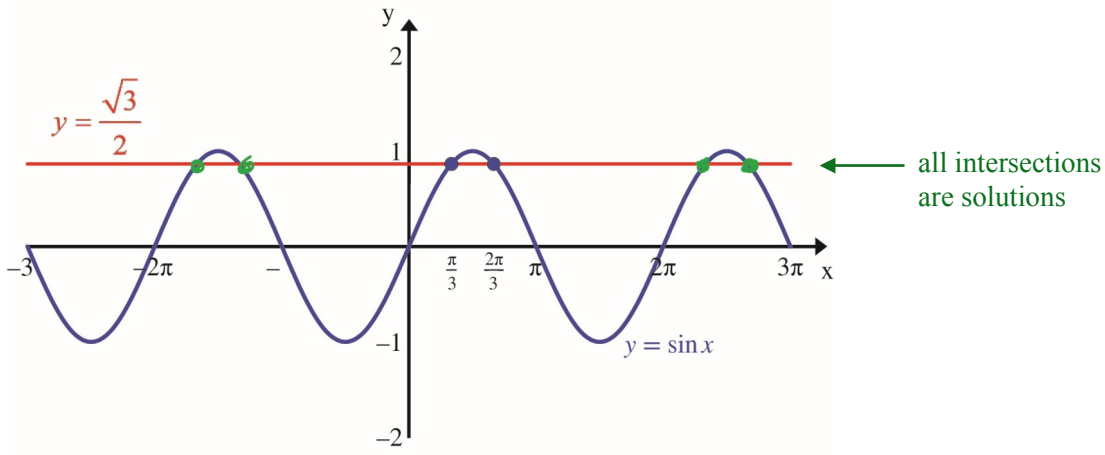
Example: Solve. $\sin x = \frac{\sqrt{3}}{2}$



From the unit circle, we can see that there are 2 values of x between 0 and 2π that will solve the equation: $x = \frac{\pi}{3}$ and $x = \frac{2\pi}{3}$.

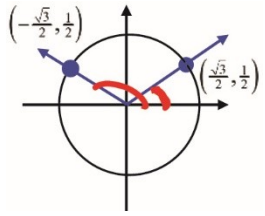
However, there are actually infinitely many solutions to the equation:

$x = \frac{\pi}{3} + 2n\pi$ and $x = \frac{2\pi}{3} + 2n\pi$ (where n is any integer).



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1. Solve the equation on the interval $[0, 2\pi)$. $3\csc x = 4 + \csc x$



$2\csc x = 4$

$\csc x = 2$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$ $\left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}$

QI QII

2. Solve the equation over the set of real numbers. $\sqrt{2}\cos x + 5 = 4$

cos-	
cos-	

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$\sqrt{2}\cos x = -$

$\cos x = -$

$x = \frac{}{}$

(

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Solve Trigonometric Equations Involving Multiple or Compound Angles

Solve the equation on the interval $[0, 2\pi)$.

3. $8\cot 2x - 5 = 3$

Isolate $\cot 2x$. $8\cot 2x = 8 \rightarrow \cot 2x = 1$

$$2x = \frac{\pi}{4} + n\pi \rightarrow x = \frac{\pi}{8} + n\frac{\pi}{2}$$

Give the general solution from the unit circle.

To find solutions on the interval $[0, 2\pi)$, substitute n = …, −1, 0, 1, 2, . . .

$n = 0$: $x = \frac{\pi}{8} + 0 \cdot \frac{\pi}{2} = \frac{\pi}{8}$

$n = 1$: $x = \frac{\pi}{8} + 1 \cdot \frac{\pi}{2} = \frac{5\pi}{8}$

$n = 2$: $x = \frac{\pi}{8} + 2 \cdot \frac{\pi}{2} = \frac{9\pi}{8}$

$n = 3$: $x = \frac{\pi}{8} + 3 \cdot \frac{\pi}{2} = \frac{13\pi}{8}$

$n = 4$: $x = \frac{\pi}{8} + 4 \cdot \frac{\pi}{2} = \frac{17\pi}{8}$ reject $> 2\pi$

4. $\cos\frac{x}{2} = \frac{\sqrt{3}}{2}$

$$\frac{x}{2} = \frac{\pi}{6} + 2\pi n, \quad \frac{x}{2} = \frac{11\pi}{6} + 2\pi n$$

$$x = \frac{\pi}{3} + 4\pi n, \quad x = \frac{11\pi}{3} + 4\pi n$$

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$n = 0: \quad x = \frac{\pi}{3} + 4\pi \cdot 0 = \frac{\pi}{3}, \quad x = \frac{11\pi}{3} + 4\pi \cdot 0 = \cancel{\frac{11\pi}{3}}$ reject $>2\pi$

$$\left\{\frac{\pi}{3}\right\}$$

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5. Find the general solution. $\sec\left(x-\frac{\pi}{4}\right)=2$

$x-\frac{\pi}{4}=\frac{\pi}{3}+2n\pi, \quad x-\frac{\pi}{4}=\frac{5\pi}{3}+2n\pi$

$x=\frac{\pi}{3}+\frac{\pi}{4}+2n\pi, \quad x=\frac{5\pi}{3}+\frac{\pi}{4}+2n\pi$

$x=\frac{7\pi}{12}+2n\pi, \quad x=\frac{23\pi}{12}+2n\pi$

$\left\{x \mid x=\frac{7\pi}{12}+2n\pi, \quad x=\frac{23\pi}{12}+2n\pi\right\}$

Solve Higher-Degree Trigonometric Equations

Solve the equation on the interval $[0,2\pi)$.

6. $2\cos^2 x-1=0$

$2\cos^2 x=1$

$\cos^2 x=\frac{1}{2}$

$\cos x=\pm\sqrt{\frac{1}{2}}$

$\cos x=\pm\frac{1}{\sqrt{2}}$

$\cos x=\frac{1}{\sqrt{2}}$ or $\cos x=-\frac{1}{\sqrt{2}}$

$x=\frac{\pi}{4},\frac{7\pi}{4}$ $\quad$ $x=\frac{3\pi}{4},\frac{5\pi}{4}$

$\left\{\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}\right\}$

7. $2\sin^2 x+5\sin x+3=0 \quad 2u^2+5u+3=0 \rightarrow (2u+3)(u+1)=0$

$(2\sin x+3)(\sin x+1)=0$

$2\sin x+3=0$ or $\sin x+1=0$

$\sin x=-\frac{3}{2}$ $\quad$ $\sin x=-1$

no solution $\quad$ $x=\frac{3\pi}{2}$ $\quad$ $\left\{\frac{3\pi}{2}\right\}$

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8. $\cos x \tan^2 x - 3\cos x = 0$

$\cos x\left(\tan^2 x - 3\right) = 0$

$\cos x = 0$ or $\tan^2 x - 3 = 0$

$x = \frac{\pi}{2}, \frac{3\pi}{2}$ $\tan^2 x = 3$

$\tan x = \pm\sqrt{3}$

The solutions $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ must be discarded as these values are not in the domain of $\tan x$.

$\tan x = \sqrt{3}$ or $\tan x = -\sqrt{3}$

$x = \frac{\pi}{3}, \frac{4\pi}{3}$ $x = \frac{2\pi}{3}, \frac{5\pi}{3}$

$\left\{\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}\right\}$

Use Identities to Solve Trigonometric Equations

Solve on the interval $[0, 2\pi)$.

9. $2\sin^2 x - 5\cos x + 5 = 0$ ← can't factor! (compare to #7)

$2\left(1 - \cos^2 x\right) - 5\cos x + 5 = 0$

$2 - 2\cos^2 x - 5\cos x + 5 = 0$

$-2\cos^2 x - 5\cos x + 7 = 0$

$2\cos^2 x + 5\cos x - 7 = 0$

$(2\cos x + 7)(\cos x - 1) = 0$

or

$2\cos x + 7 = 0$

$\cos x = -\frac{7}{2}$

no solution

$\cos x - 1 = 0$

$\cos x = 1$

$x = 0$

$\{0\}$

10. $\sin 2x = \cos x$

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$2\sin x\cos x = \cos x$

$2\sin x\cos x - \cos x = 0$

$\cos x(2\sin x - 1) = 0$

$\cos x = 0 \quad \text{or} \quad 2\sin x - 1 = 0$

$x = \frac{\pi}{2}, \frac{3\pi}{2} \qquad \sin x = \frac{1}{2}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$

$\left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}\right\}$

11. $\sin x = \cos x$

$\sin x - \cos x = 0$

$(\sin x - \cos x)^2 = 0$

$\sin^2 x - 2\sin x\cos x + \cos^2 x = 0$

$1 - 2\sin x\cos x = 0$

$1 - \sin 2x = 0$

$\sin 2x = 1$

$2x = \frac{\pi}{2} + 2n\pi \rightarrow x = \frac{\pi}{4} + n\pi$

$n = 0,\ x = \frac{\pi}{4} \qquad n = 1,\ x = \frac{5\pi}{4}$

$\left\{\frac{\pi}{4}, \frac{5\pi}{4}\right\}$

Both answers check.

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12. $\sin x = \cos x$ (alternate solution) Note that $\cos x = 0$ will not solve the equation.

$$\frac{\sin x}{\cos x} = 1, \quad \cos x \neq 0$$
$$\tan x = 1$$
$$x = \frac{\pi}{4}, \frac{5\pi}{4} \quad \left\{\frac{\pi}{4}, \frac{5\pi}{4}\right\}$$

13. $\sin x - 1 = \cos x$

$$(\sin x - 1)^2 = \cos^2 x$$
$$\sin^2 x - 2\sin x + 1 = \cos^2 x$$
$$\sin^2 x - 2\sin x + 1 = 1 - \sin^2 x$$
$$2\sin^2 x - 2\sin x = 0$$
$$2\sin x\,(\sin x - 1) = 0$$
$$\sin x = 0 \text{ or } \sin x = 1$$
$$x = 0, \pi \qquad x = \frac{\pi}{2}$$

Check! $\sin x - 1 = \cos x$

$x = 0$: $0 - 1 = 1$ No

$x = \pi$: $0 - 1 = -1$ Yes

$x = \frac{\pi}{2}$: $1 - 1 = 0$ Yes

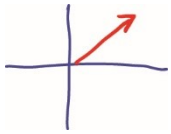
$$\left\{\pi, \frac{\pi}{2}\right\}$$

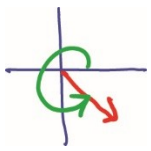
Use Inverse Functions to Express Solutions to Trigonometric Equations

14. Solve on the interval $[0, 2\pi)$. $4\cos^2 x - 7\cos x + 3 = 0$

$$(4\cos x - 3)(\cos x - 1) = 0$$
$$\cos x = \frac{3}{4} \text{ or } \cos x = 1$$
$$x = 0$$
$$x = \cos^{-1}\frac{3}{4} \approx 0.7227$$

$$\left\{0, \cos^{-1}\frac{3}{4}, 2\pi - \cos^{-1}\frac{3}{4}\right\}$$





$$x = 2\pi - \cos^{-1}\frac{3}{4}$$
$$\approx 5.5605$$

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Approximate the Solutions to Trigonometric Equations on the Calculator

15. Use a graphing utility to solve. Round the solution to 4 decimal places.

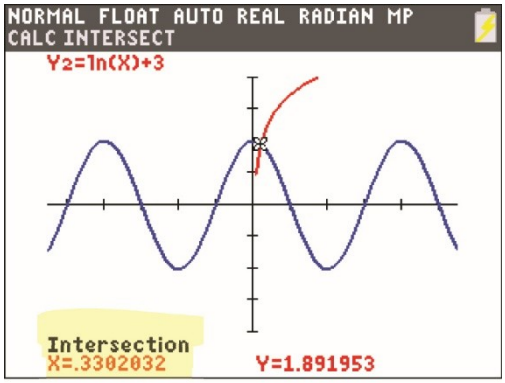
$2\cos x = \ln x + 3$

$Y_1 = 2\cos x,\ Y_2 = \ln x + 3$

window: zoom trig

use: intersection $x \approx 0.3302$

$\{0.3302\}$



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